

19 – Kernel Services

Design of the platform kernel (layer 0 of doc 17): the five services designed in full here – identity and authentication (OpenID Connect), the permission catalogue and role builder, the audit service, internationalisation on ICU, and the event model – plus the index to the kernel services designed in docs 18 and 32, and the three short specifications that had no home: notifications delivery, the configuration register, and scheduling.

Field	Value
Document	19 – Kernel Services (design artefact; Stage 1; renamed from Cross-Cutting Services in v0.2 to match doc 17's layer-0 vocabulary)
Version	v0.2 – draft for design review (rename; Section 0 kernel index; Sections 7–9 notifications delivery, configuration register, scheduling; invariants and traceability extended)
Date	8 September 2026
Status	Draft. Depends on doc 18 tables (app_user, role, permission, user_role, audit_event, device, numbering_series). Broker and identity-provider product choices are left as decision requests (Section 8).
Inputs	05 SRS (FR-SEC, FR-GOV, FR-LOC, FR-REP), 06 NFR Specification, 07 Matrices (roles, audit event catalogue, configuration register), 09 Decisions Log (ADR-10, 18, 24, 26, 27), 16 Questionnaire Answers (sections H, M, N, O, P, Q, R), 18 Core Data Model, and the September 2026 review decisions.
Audience	Engineering leads. Sections 2 and 8 are the parts a business reader needs.
Scope boundary	Service behaviour, contracts and rules. Product selection (identity provider, message broker) is framed with criteria and a default, not decided; the sync <i>protocol</i> (message formats, batching, retry timing) is a separate document.

Why these five are designed together. A permission is meaningless without an identity to hold it; an audit event is meaningless without both; every audit event and every screen string must render in three languages; and all of it moves through one event model. Designing them apart produces five inconsistent notions of “who”, “where” and “when”. Designing them together lets one set of claims (user, device, entity scope, location scope, correlation) flow unchanged from login to audit row to report.

0. The kernel: what it is and where each service is designed

Doc 17 (section 6) places ten technical services in layer 0, the platform kernel: services with no business rules that every module may use and that depend on nothing above them. This document is the complete index. Five are designed in full here; three are designed in the documents that own their tables or protocol; three had only a sketch in doc 17 and receive their specification in Sections 7–9 of this version.

Kernel service	Responsibility (doc 17 §7.11)	Designed in	Implemented by
Scope and session	Set the caller's entity/location scope and policy class on every request and database session; RLS policy classes	This doc §1, §3.4; doc 18 §3.7	19A
Identity and authentication	OIDC provider, tokens, TOTP step-up, till PIN, device credentials	This doc §2	19A
Permission catalogue and role builder	Permissions as data, templates, segregation of duties, till snapshot	This doc §3; doc 18 sod_pair	19A (service), 21/21A (data and screens)
Audit	Append-only store through one facade in the business transaction; exception queues; hash chain P2	This doc §4; doc 18 part D	19A
Event backbone	Outbox archive, relay, broker, inbox, replay; envelope	This doc §6; doc 18 part E	19A
Internationalisation	ICU collation, NFC, message catalogue with English fallback, rendering pipelines, input-method register	This doc §5	19A (plumbing), 17A (shell fonts and catalogue loading)
Documents & Numbering base	Document header, lines, links, state history, attachments; document-type registry; series and issuance protocol; gap detection	Doc 18 part C (§5); numbering addendum 24B after the tax advisor	19A
Edge sync gateway	Upload, distribution, heartbeat, series registration, counter transfer, rate limiting	Doc 32; doc 18 part E	19A (gateway), 26A (till side)
Attachments	Object-storage keys and metadata, pre-signed upload, hash verification	Doc 18 document_attachment; doc 32 §4	19A
Notifications delivery	Event-triggered sends through M9's adapter and templates, with retry	This doc §7	19A (delivery), 29A (templates, adapter)
Configuration register	Onboarding-time values with defaults, scoped per entity or location, audited	This doc §8 ; doc 07 §4 (the register content)	19A
Scheduling	Singleton jobs under cluster locks; job catalogue; run history	This doc §9 ; doc 14 §1.3	19A

The rule that keeps the kernel small: a service belongs here only if it carries no business rule. Price resolution is *not* kernel (it is M3's shared engine library, ADR-31); document lifecycles are *not* kernel (the base is; the state machines belong to M4, M5 and M6).

1. Overview and shared vocabulary

The five services share one identity vocabulary. It is established at authentication, carried in the access token, set on every database session, stamped on every event and audit row, and used by every report. No service invents its own.

Claim	Source	Carried in	Used by
sub` (user id)	Identity service	Token, DB session, events, audit	Attribution (H-13), RLS scope
dev` (device id)	Device enrolment	Device token, events, audit	Series holder, sync sequence, fleet health
ent` (home entity)	app_user	Token, DB session	P1 ownership, RLS OWN class
scopes` (entity/location pairs)	user_role	Token	RLS scope, till mode gating
roles` + `rv` (role set, catalogue version)	user_role, permission catalogue	Token	Local permission resolution, cache invalidation
cls` (policy class)	Role class	Token, DB session	RLS: OWN / PARTY / FEDERATION_VIEW / EXTERNAL_TIMEBOXED
mfa_at	Identity service	Token	Step-up freshness for requires_mfa permissions
corr` (correlation id)	First request in an action	Every event and audit row	Grouping one user action end to end
lang	Location language	Token, request header	Rendering; never affects data

Interaction in one sentence: the **identity service** authenticates and mints tokens; the **permission service** resolves what those tokens may do; the **audit service** records what they did; the **i18n service** renders it in the reader's language; the **event model** moves all of it between modules, tills, central and reports.

2. Identity and authentication (OIDC)

2.1 Principals

Four kinds of principal exist, and they authenticate differently. Conflating them is the classic mistake; keeping them distinct is what allows a till to work offline for two days while a back-office finance function demands a fresh second factor.

Principal	Authenticates with	Token	Notes
Back-office user	Username + password at the central OpenID Connect (OIDC) provider; TOTP second factor for functions marked requires_mfa (N-04)	OIDC access token (JWT, 15 min) + rotating refresh token (8 h, single-use)	NIST SP 800-63B credential rules: minimum 12 characters, breached-password screening, no forced periodic rotation, rate-limited attempts. Idle lock 10–15 min; admin screens 5 min (N-09).
Till operator	Short numeric PIN, verified <i>on the device</i> against a cached Argon2id hash (N-03)	Local operator session on the device; no central token	PIN is a low-entropy credential and is acceptable only because it is bound to device possession, rate-limited (5 attempts then 15-minute lockout), and scoped to till permissions. 4–6 digits, entity policy. Idle lock 2–3 min.
Device	Enrolment credential: an X.509 client certificate (or client secret where certificates are impractical) bound to hardware serial	OAuth2 client-credentials device token (24 h, refreshed on sync)	The device token authenticates sync uploads and downloads. Suspension is enforced at central immediately and on the device at next contact; the device wipes its caches on a signed revoke.
External system or person	API consumer: OAuth2 client credentials with entity scope. Regulator/auditor: username + password + mandatory TOTP	Access token with `cls` = EXTERNAL_TIMEBOXED` and `exp` = grant expiry	Every grant and every login is audited (M-10, J-09). Single sign-on with external identity providers is out of scope.

2.2 Provider and topology

One central OIDC provider serves the whole federation as a **single realm**. Entity is a claim, not a realm: users from 876 entities share one login surface, and cross-entity principals (Federation view, auditors, regulators) are ordinary users with a different policy class. Per-entity realms were considered and rejected: they would multiply configuration 876-fold and make the Federation view a federation-of-realms problem.

Requirement	Design
Delegated administration (M-08, ADR-18)	Entity administrators manage users whose home entity is theirs; an MPCS administrator additionally manages users scoped to its shops. The Federation holds no downward administrative capability: the provider's admin API is wrapped by the platform's own user-management service, which enforces the scope rule, and no Federation role maps to the provider's native administrator. The consequence is accepted and documented: the helpdesk cannot reset an MPCS user's password (Q-09).
Password reset	Entity administrator issues a one-time temporary password (audited USER_CREDENTIAL_RESET); self-service reset by SMS one-time code to the registered phone is phase 2 (R-08 notifications).
Step-up authentication	Permissions with requires_mfa trigger a TOTP challenge unless `mfa_at` is within the freshness window (default 12 hours; per-entity configurable down to per-action). The challenge is at the action, not at login, so ordinary work is not slowed.
Token content	Claims in Section 1. Permissions are not in the token (a role can hold hundreds); role ids plus catalogue version `rv` are, and the permission service resolves locally against its cached catalogue.
Revocation	Access tokens are short-lived and not revoked; refresh tokens are single-use and revocable; user deactivation revokes all refresh tokens and device-cached PIN hashes are removed at next sync.
Offline tills	Device holds: device credential, cached PIN hashes for operators assigned to the shop, cached role-to-permission snapshot with version, and cached `offline_allowed` flags. A user deactivated at central keeps working on an offline till until the next sync – an accepted consequence of H-01, mitigated by the shop administrator's ability to remove the operator locally (audited).
Provider product	Decision request DR-1: a standards-compliant self-hosted provider (Keycloak-class) versus Spring Authorization Server embedded in the monolith. Criteria in Section 8.

2.3 Sessions and locks

Surface	Idle lock	Absolute session	Re-authentication
Till (sales mode)	2–3 min (configurable per entity)	Till day-close	PIN
Till (shop-management mode)	3 min	Mode ends on return to sales	PIN of a user holding shop-management permissions; mode switch audited (MODE_SWITCH_SHOP_MGMT)
Back office	10–15 min	8 h refresh lifetime	Password; TOTP on step-up
Administration screens	5 min	8 h	Password + TOTP always
External time-boxed	5 min	Grant expiry	Password + TOTP always

3. Permission catalogue and role builder

3.1 The catalogue

Permissions are data (doc 18, table permission). Each is a fixed string of the form `module.resource.action`; the catalogue is versioned as a whole, and the version travels in the token as `r.v.` A representative extract:

Permission code	offline	mfa	scope	Meaning
pos.receipt.issue	yes	no	LOCATION	Issue a till receipt
pos.receipt.void	yes	no	LOCATION	Void within same session
pos.refund.same_session	yes	no	LOCATION	Refund on same till, same session
pos.refund.cross_session	no	no	LOCATION	Refund across tills or days (supervisor, online)
pos.negative_stock.acknowledge	yes	no	LOCATION	Proceed with a sale the records say is impossible (E-09)
pos.mrp.pick	yes	no	LOCATION	Use the batch picker where policy is PICKER
shop.grn.confirm	yes	no	LOCATION	Confirm a goods received note; ownership transfers
shop.count.record	yes	no	LOCATION	Record a stock count
inv.writeoff.request	yes (draft)	no	LOCATION	Raise a write-off draft
inv.writeoff.witness	no	no	ENTITY	Witness a write-off (distinct person)
inv.writeoff.approve	no	yes	ENTITY	Approve; carries value limit parameter
inv.adjust.approve	no	yes	ENTITY	Approve count adjustment; value thresholds by SKU or tag (E-07)
ord.order.submit	no	no	ENTITY	Raise an order to a seller
ord.order.accept	no	no	ENTITY	Accept, allocate, set ETA (seller side)
bil.invoice.issue	no	no	ENTITY	Issue a tax invoice
bil.creditnote.issue	no	yes	ENTITY	Issue a credit note
bil.creditlimit.change	no	yes	ENTITY	Change a relationship credit limit (A-11)
prc.pricelist.author	no	no	ENTITY	Author retail or trade price lists
prc.rule.author	no	no	ENTITY	Author discount and markdown rules
prc.controlprice.enter	no	yes	FEDERATION	Enter a gazetted control price
cat.sku.create_local	no	no	ENTITY	Create a local SKU
cat.sku.promote	no	no	FEDERATION	Promote local to shared
gov.user.manage	no	yes	ENTITY	Create, deactivate, reset users in scope
gov.role.manage	no	yes	ENTITY	Build and assign roles in scope
gov.external.grant	no	yes	FEDERATION	Grant a time-boxed external view
gov.audit.review	no	no	ENTITY	Acknowledge exception-report items
sys.device.enrol	no	yes	ENTITY	Enrol a device; assign to a position

Three attributes do most of the work. `offline_allowed` is exactly what the till caches (H-02/H-03). `requires_mfa` marks the online-only financial and administrative set (N-04). `scope` states the finest scope at which the permission can be granted: a LOCATION permission granted at entity scope covers all the entity's locations; a FEDERATION permission cannot be granted by anyone but the Federation.

3.2 Segregation of duties

Two enforcement modes exist, chosen per conflict pair, because the canonical rule (a write-off requester may not approve it, N-07) is about the *instance*, not about holding both permissions. A one-person MPCS accounts office may legitimately hold both.

Conflict pair	Default mode	Effect
writeoff.request ↔ writeoff.approve	INSTANCE	Same user may hold both; may not approve a write-off they requested. Enforced by comparing actor to the document's requester at approval.

Conflict pair	Default mode	Effect
writeoff.request ↔ writeoff.witness	INSTANCE	Witness must differ from requester (F-05). At a single-staff shop the MPCs approver fulfils the witness role remotely with mandatory photographs (September 2026 decision, DR-7 of doc 18).
adjust.request ↔ adjust.approve	INSTANCE	As write-off.
user.manage ↔ audit.review (own actions)	INSTANCE	A reviewer may not acknowledge exception items where they are the actor.
role.manage ↔ (any)	GUARDRAIL	Nobody can grant a permission they do not themselves hold; nobody can remove the last user holding gov.user.manage in an entity.

Entities may raise a pair from INSTANCE to ROLE mode (the two permissions can no longer coexist in one role for that entity). The change is a configuration-register item and is audited.

3.3 Role builder

Aspect	Design
Templates	Federation-owned roles with <code>`is_template = true`</code> : the fourteen of doc 07 plus shop-in-charge (cashier + receiving + count + write-off request, for single-staff shops), driver (proof-of-delivery capture only) and the two external classes (regulator view, auditor view).
Adoption	An entity clones a template; the clone is its own role from that moment. Template changes do not propagate automatically (entity autonomy); the entity administrator sees a "template updated" marker and may re-sync with a diff shown. Decision request DR-4 confirms this against the alternative of forced propagation for security-critical changes.
Parameters	Some permissions carry limits: approval value ceilings, variance thresholds. These live as <code>`limits jsonb`</code> on <code>role_permission</code> , validated against a per-permission parameter schema, so a role can say "approve write-offs up to Rs 25,000" without a new permission code.
Guardrails	No privilege escalation (grantor must hold the permission); FEDERATION-scope permissions never appear in entity role builders; a role cannot be deleted while assigned; every change is a <code>ROLE_CHANGE</code> audit event with before/after permission sets.
Versioning	Each role carries a version; the catalogue carries <code>`rv`</code> . A token whose <code>`rv`</code> is behind is resolved against the current catalogue (permissions can only be more restrictive at runtime, never more permissive than the current catalogue allows).
Till snapshot	The till caches, per operator assigned to the shop, the resolved permission set filtered to <code>`offline_allowed`</code> , with <code>`rv`</code> . Snapshot refresh is part of every sync; the snapshot is signed so a tampered cache fails closed.

3.4 Enforcement points

Point	What is checked	Mechanism
API and service layer	Permission code required by the operation; step-up freshness where <code>requires_mfa</code>	Declarative annotation per service method mapping to permission codes; a method without an annotation fails the build (a static test walks the service layer).
Database	Row scope only (never permissions)	RLS policy classes of doc 18 section 3.7, keyed on session variables set from the token at connection checkout.
Till	Permission from the cached snapshot; <code>offline_allowed</code> when offline	Same permission codes, resolved locally; the till never carries permissions it did not receive in a signed snapshot.
User interface	Visibility only	Screens hide what the user cannot do; this is convenience, never security.

4. Audit service

Doc 18 defined the audit tables; this section defines the service that fills and reads them. The design goal is that an auditor can trust the log without trusting the application: structure, not procedure, guarantees completeness and immutability.

4.1 Emission

Rule	Design
Same transaction	The audit row is inserted in the same database transaction as the business change it records, through one audit facade. There is no separate audit queue at central; a business change without its audit row cannot commit. On the till the same holds against the local database.
Facade, not free text	Code calls <code>`audit.record(eventType, subject, before, after, reason, witness)`</code> . Event types are the catalogue of doc 18 (<code>audit_event_type`</code>); a code path cannot invent a type, and every REVIEW/ALERT type must have a reviewer template before deployment (D-I3).
Correlation	The first request in a user action creates <code>`corr`</code> ; it propagates through service calls, the outbox and the till sync so that a refund appears as one action (REFUND, DRAWER_OPEN, state transition) rather than three unrelated rows.
Minimal diff	<code>before_state`</code> / <code>`after_state`</code> carry only changed columns. For immutable subjects (documents) <code>`before_state`</code> is null and the event points at the document; the document is its own record.
Never logged	Credential material (passwords, PIN hashes, TOTP secrets, tokens), full card references, and personal data values. A customer phone change logs the customer id and the fact; the values live under PDPA rules in the customer table (ADR-27).
Reason capture	Where the type demands a reason (voids, refunds, write-offs, credit-limit changes, negative-stock acknowledgements) the user interface presents a coded reason list in the shop language plus optional free text; the code is what reports aggregate on.

4.2 Immutability and integrity

Layer	Phase	Design
Privileges	1	Application role: INSERT only on <code>audit_event`</code> ; no UPDATE, DELETE, TRUNCATE; no ownership of the table. A deployment test asserts the grant set (D-I1). Partitions are detached to cold storage, never dropped.
Device sequence	1	Every till-originated event carries <code>`(device_id, device_seq)`</code> , dense per device. Central raises SYNC_ANOMALY on any gap or duplicate that is not an idempotent replay.
Hash chain	2	Per monthly partition: <code>`row_hash = SHA-256(prev_hash    canonical(row))`</code> . The chain head for each closed partition (a daily anchor for the open one) is written to write-once object storage under a separate credential the application does not hold. Verification recomputes the chain and compares to the anchor; a mismatch is an ALERT.
Export	2	Security information and event management (SIEM) export of ALERT events over a push channel; not a phase-1 dependency.

4.3 Review: exception reports that someone owns

An exception report nobody owns is a control that does not exist (N-06). The service therefore ties every REVIEW and ALERT type to a reviewer role template and tracks acknowledgement as a further audit event, never as a status flag on the original.

Element	Design
Queues	One exception queue per entity, materialised from REVIEW/ALERT events in scope, with the per-entity freshness stamp (O-03) so the reviewer knows which shops have not synced.
Acknowledgement	<code>gov.audit.review`</code> raises AUDIT_REVIEWED with <code>`subject_id`</code> = the reviewed event and a disposition code (NOTED, INVESTIGATE, CLOSED_WITH_ACTION). Self-review is blocked (Section 3.2).
Escalation	Unacknowledged ALERT items older than a configured age (default 3 days) escalate to the next role in the entity's escalation list; Federation sees all queues view-only (M-04).
Regulator and auditor reads	Entity-scoped or federation-wide SELECT through the same RLS classes as operational data; every access session is itself an audit event.

5. Internationalisation on ICU

Trilingual is a phase-1 requirement across interface, printed output and search (P-01, P-04). The design separates four concerns that are often conflated: **interface strings** (a translation catalogue owned by the product team), **master data** (three columns per name, owned by the catalogue steward), **rendering** (fonts, shaping, formatting), and **collation and search** (how Sinhala and Tamil sort and match).

5.1 Language resolution

Rule	Design
Where language comes from	Per shop (P-02): <code>`location.language`</code> , inherited by every device at that location and every user session opened there; back-office users take the language of their office location, falling back to <code>`entity.default_language`</code> . There is no per-transaction switch in phase 1.
Fallback	Field-level, never document-level: a receipt in Tamil with one SKU lacking a Tamil name prints that line in English and the rest in Tamil. English is the mandatory anchor (P-07); a missing translation is never rendered as blank or as a key.
Authoritative text	English, for legal documents and disputes (P-07). Printed tax invoices carry the English name alongside the shop-language name.
Numerals and dates	Western Arabic digits (0–9) in every language, including Sinhala and Tamil interfaces; dates <code>`dd/mm/yyyy`</code> ; currency rendered <code>`Rs 1,234.00`</code> . ICU locale data for <code>`si-LK`</code> and <code>`ta-LK`</code> would default to native numerals for some formats; the platform overrides this once, centrally (decision request DR-6).

5.2 Interface strings

Element	Design
Catalogue	One message catalogue keyed by stable message ids (<code>`pos.receipt.total`</code> , not English text), with <code>`en`</code> , <code>`si`</code> , <code>`ta`</code> entries; ICU MessageFormat for plurals and placeholders. Shipped with each release; the till holds it locally.
Ownership	Product team owns the catalogue; a translation review by a named Sinhala and Tamil reviewer is a release gate, and a release cannot ship with untranslated ids (a build test counts them).
Screens as paper	The usability directive (P-08): screens mirror the paper documents shop staff already know; message ids therefore follow document vocabulary (GRN, receipt, count) not software vocabulary.

5.3 Rendering

Surface	Design
Till interface (Android)	Platform text stack with Noto Sans Sinhala and Noto Sans Tamil bundled in the application (do not rely on device fonts). Complex-script shaping is native on Android.
Thermal receipts	Rendered as a raster bitmap by the till's text stack and sent to the printer as an image (P-05). This sidesteps the printer's own font support, which is the usual cause of broken Sinhala and Tamil on low-cost printers. Golden receipts in all three languages form the acceptance test, including a mixed-language line (English fallback inside a Tamil receipt).
A4 documents and reports	Server-side rendering with embedded Noto fonts and a HarfBuzz-based shaping engine; the same pipeline renders every catalogue report and printed document (P-03).
Web back office	Web fonts for the three scripts served with the application; browser shaping is adequate.
Input	Sinhala and Tamil input methods provisioned per shop at onboarding (P-06); the application accepts any Unicode input and normalises to NFC on write (Section 5.4).

5.4 Collation, normalisation and search

Rule	Design
Normalisation	All text is normalised to Unicode NFC at the write boundary. Sinhala and Tamil have multiple encodings for visually identical text; without normalisation two spellings of the same name will neither sort together nor match.
Collation	PostgreSQL ICU collations: <code>`si-x-icu`</code> on <code>`_si`</code> columns, <code>`ta-x-icu`</code> on <code>`_ta`</code> , <code>`en-x-icu`</code> on <code>`_en`</code> . Sort order in every list follows the display language's collation (P-04, P-09).
Search at the till	Prefix and partial match on all three name columns simultaneously (the cashier may type in any script), using trigram indexes per column; results ordered by the display language's collation. Product-code lookup remains the fast path for unbarcoded items (B-11).
Search in the back office	Same in phase 1; OpenSearch-backed analysers with script-aware tokenisation are phase 2 and index from the same columns.

Rule	Design
Test set	A curated list of Sinhala and Tamil product and place names with known correct sort order and known equivalent spellings is a repository asset, run on every build.

6. Event model

Four things are called “events” in this system and they are not the same thing. Naming them precisely is half the design.

Kind	Purpose	Producer	Consumers	Persistence
Domain event	Tell other modules that a fact changed	Any module, inside its transaction	Reporting projections, notifications, sync push, integrations	Outbox table (the replayable log)
Sync record	Move till-originated facts to central, and central-published reference data to tills	Till outbox; central publisher	Central ingest; till applier	Till outbox until acknowledged; central inbox for idempotency
Audit event	Record who did what	Audit facade	Exception queues, auditors	audit_event (doc 18), indefinitely
Analytics event	Feed near-real-time dashboards and end-of-day aggregates (O-03)	Projection from domain events	Reporting store	Reporting store; rebuildable from the outbox

6.1 Envelope

One envelope for all four kinds, so tooling, ordering and idempotency are written once:

Field	Type	Rule
event_id	uuid (v7)	Idempotency key. Generated by the producer.
event_type	text	Versioned name, e.g. `document.issued.v1`, `sku.promoted.v1`, `control_price.entered.v1`. A payload change is a new version; consumers declare which versions they accept.
occurred_at, occurred_local	timestampz	P9 of doc 18.
source	text	central` or the device id.
source_seq	bigint	Dense per source. The ordering key; clocks never order events.
owner_entity_id, location_id	uuid	Scope; used for routing (which tills receive it) and for RLS on projections.
aggregate_type, aggregate_id	text, uuid	What changed.
corr, causation_id	uuid	Correlation with the audit log; the event that caused this one.
actor	uuid	User or system actor.
payload	jsonb	Type-specific. Small: identifiers and changed facts, not whole documents. Consumers fetch by id when they need more.

6.2 Outbox and delivery

Element	Design
Transactional outbox	A module writes its domain event into `event_outbox` in the same transaction as the change. Nothing is published that did not commit; nothing commits without its event. The outbox is retained (partitioned monthly) and is the replay source for rebuilding any projection.
Relay	A relay process reads the outbox in `source_seq` order and publishes to the broker, marking rows published. At-least-once delivery; consumers are idempotent by `event_id` (an `event_inbox` table per consumer records applied ids).
Broker	Needed for fan-out and buffering, not for retention (the outbox retains). This rules out needing a log-retention broker in phase 1; a persistent-queue broker with consumer groups and dead-letter queues suffices. Decision request DR-2 selects between RabbitMQ (quorum queues) and NATS JetStream; Kafka is held for a later phase if an analytics workload demands log replay at the broker rather than from the outbox.
Ordering	Guaranteed per source (a device, or central); not guaranteed across sources. Consumers that need cross-source order (federation aggregates) order by `received_at` and stamp freshness (O-03).
Dead letters	A poison event goes to a dead-letter queue after three attempts and raises an ALERT audit event; it never blocks the source's subsequent events beyond the retry window.

Element	Design
Burst tolerance	After a wide outage thousands of tills reconnect with up to two days of records each (H-01). Ingest is rate-limited per device and prioritises documents and movements over audit rows within each device's sequence window; the broker absorbs the burst; projections lag rather than fail, and the freshness stamp says so.

6.3 The sync record in this model

The sync protocol is its own document; here is only its place in the event model, following ADR-10 (one writer per record type).

Direction	Content	Rule
Till → central	Issued documents with lines and state rows; stock movements; audit events – all as envelope events in `source_seq` order	Central verifies content hashes and sequence density, applies idempotently, acknowledges the last applied `source_seq`. Central never re-numbers, never edits, never merges.
Central → till	Catalogue, barcodes, batches and lots for the location, price lists, discount rules, control prices, permission snapshot, message catalogue – as versioned snapshots plus incremental change events routed by location	Tills apply read-only replacements. A till applies a snapshot version atomically; it never runs on a half-applied catalogue. Control-price and rule changes carry an `apply_from` so all tills switch on the same business date.
Conflict	None	By construction: no record type has two writers.

6.4 Core domain event catalogue (phase 1)

Event type	Emitted when	Main consumers
entity.*`, `location.*`, `device.*	Party and fleet changes	Sync routing, reporting
sku.created/updated/promoted`, `barcode.registered`, `batch.created	Catalogue changes	Till snapshot push, reporting
price_list.published`, `discount_rule.published`, `control_price.entered/rescinded	Pricing changes	Till push with `apply_from`; reporting
document.issued`, `document.state_changed`, `document.linked	Any document lifecycle step	Stock projection, AR/AP picture, notifications (dispatch alerts, statements), reporting
stock.moved	Every movement	Balances, ageing, expiry warnings, reorder suggestions (P2)
user.*`, `role.changed`, `external_grant.*	Governance changes	Permission snapshot push, audit review
sync.batch_received`, `sync.anomaly	Ingest outcomes	Freshness stamps, exception queues, fleet health

6.5 Reporting projections

Near-real-time operational dashboards read projections built from domain events; heavy aggregates run end of day (O-03). Phase 1 keeps projections in a read replica of the primary database, provider-neutral. A columnar analytics store is introduced when projection volume demands it (decision request DR-3 sets the trigger), and because the outbox is the replay source, that migration is a rebuild, not a migration.

7. Notifications delivery

The kernel delivers; M9 owns what is said and through which provider (doc 20 M9). Keeping delivery in the kernel means every module raises a notification the same way – by publishing a domain event – and none calls a gateway directly.

Aspect	Design
Trigger	A notification rule maps an event type (with an optional predicate on payload fields) to a template and an audience rule. Rules are data owned by M9; the kernel evaluates them as an ordinary event consumer.
Audience	Resolved at send time from M1: named roles at the counterparty entity (dispatch alert to the buyer's stores role), the customer's phone (statement), or an explicit recipient on the event.
Rendering	Template in the recipient's language with English fallback, through the i18n service; placeholders from the event payload plus a bounded lookup (document number, amounts, dates) – never a query into another module's tables.
Channels	SMS and e-mail in phase 1 through M9's provider adapter interface; in-app (the exception queue and a notification bell) is a third channel with no external provider.
Delivery record	notification_log: id, rule, event id, recipient (hashed for SMS numbers), channel, rendered length, status (QUEUED, SENT, FAILED, SUPPRESSED), provider reference, attempts. Insert-plus-status-update only.
Retry and failover	Three attempts with backoff; then the secondary provider if configured; then FAILED with an ALERT audit event. Never retried past 24 hours (a dispatch alert two days late is noise).
Suppression	Per recipient quiet hours (22:00–07:00 default for SMS), de-duplication of identical notifications within an hour, and an entity-level kill switch for a channel.
Privacy	Message bodies are not stored beyond the rendered length; phone numbers are hashed in the log; the audit log records the fact of sending, not the content (ADR-27).
Guarantee	At-least-once toward the provider; idempotent by (rule, event id, recipient) so a replayed event does not resend.

8. Configuration register

Doc 07 section 4 lists what is configurable; each module design section 7 adds its items. The kernel holds them all in one place with one resolution rule, so no module keeps private settings.

Aspect	Design
Model	config_item (key, value type, validation schema, description in three languages, default, scope kind: FEDERATION ENTITY LOCATION, who may change: permission code) and config_value (key, scope entity, scope location nullable, value jsonb, effective_from, changed_by, changed_at). Values are append-only; the current value is the latest effective row.
Resolution	Most specific scope wins: location → entity → federation default. A location value exists only for keys whose scope kind allows it. Resolution is a kernel query cached per instance and invalidated by config.changed.v1.
Validation	Every value is validated against the item's schema at write; a key nobody registered cannot be set. Law-like items (control prices, segregation pairs) are not configuration items – they are data with their own tables (P10 of doc 17).
Distribution	Items flagged till_visible are part of the shop snapshot (doc 32 §5.1): idle locks, quiet hours, offline account cap, multi-MRP thresholds, staleness age.
Audit	Every change raises config.changed.v1 with before/after; changes to items flagged sensitive (credit-related, security) require MFA.
Onboarding	The onboarding playbook (doc 37) sets location-scoped values from the checklist; BulkRegister (doc 21) accepts a configuration sheet for the same keys.
Phase-1 catalogue seed	The doc 07 register plus the items named in docs 19 §8, 21 §7 and the later module designs; the seed file lives in the repository and is the single source (19A).

9. Scheduling

Singleton work must run exactly once across N stateless instances (doc 14 section 1.3, P12). The kernel provides the lock, the catalogue and the run history; modules provide the job body.

Aspect	Design
Job catalogue	scheduled_job (name, owning module, schedule expression, timezone Asia/Colombo, lock timeout, max runtime, enabled). Registered in code; the table is the runtime view and the on/off switch.

Aspect	Design
Locking	A database advisory lock (or ShedLock-class row lock) per job name acquired for the run; a second instance skips silently. Lock timeout longer than the job's maximum runtime; a job exceeding it is killed and reported, never overlapped.
Run history	job_run (job, started_at, finished_at, instance, outcome, items processed, error). Insert-plus-finish only; feeds the fleet and health views.
Phase-1 jobs	Outbox relay (continuous, partitioned by source); end-of-day aggregation per business date; expiry-window scan; statement generation; stock reconciliation (B-I4); series gap check (C-I2, E-I2); audit anchor (P2); snapshot rebuild and change-log pruning; ring promotion (doc 31); idempotency-key expiry; notification retry sweep.
Business date	Jobs keyed to a trading day run per location day-close where relevant (end-of-day), not at a fixed clock time for all shops; the scheduler exposes a day-close trigger from M6's session close events.
Failure	A failed run is retried on the next schedule, not immediately, unless flagged critical (relay, gap check); repeated failure raises an ALERT.
Manual runs	Permission-gated (sys.job.run, MFA) from the fleet console; audited.

10. Service invariants

The contract that code, configuration and tests must all honour.

- X-I1. No request executes without a resolved principal and scope; anonymous access exists only for the login endpoints.
- X-I2. Permissions are checked at the service layer by code, never inferred from the user interface; every service method carries a permission annotation (build-time test).
- X-I3. No user can grant a permission they do not hold; no entity can lose its last user-management holder.
- X-I4. A requires_mfa permission is exercised only with `mfa\_at` inside the freshness window.
- X-I5. A till exercises only permissions present in its signed snapshot; offline, only those flagged offline_allowed.
- X-I6. Every business change commits with its audit row and its domain event in the same transaction, or not at all.
- X-I7. Every event and audit row carries the correlation id of the user action that caused it.
- X-I8. Events from one source are applied in `source\_seq` order, exactly once in effect (idempotent by `event\_id`).
- X-I9. Every string shown to a user resolves through the message catalogue or a three-column name with English fallback; a raw message id or a blank is a defect.
- X-I10. All stored text is NFC-normalised; all list ordering uses the display language's ICU collation.
- X-I11. No module calls a notification, SMS or e-mail gateway directly; every notification originates from a domain event through a registered rule.
- X-I12. No module holds private settings; every configurable value resolves through the configuration register with a registered key and schema.
- X-I13. Every scheduled job runs under the kernel lock; two instances never execute one job concurrently.

11. Decision requests

#	Point	Default proposed	Criteria / consequence
DR-1	Identity provider product	Self-hosted standards-compliant provider (Keycloak-class)	Must support: single realm with attribute-based delegation, TOTP, client-certificate device auth, admin API wrappable behind the scope rule, operable by one DevOps engineer. Embedded Spring Authorization Server is the fallback if the provider's operational weight is judged too high.
DR-2	Message broker	RabbitMQ with quorum queues	Managed offering on DigitalOcean and installable locally; consumer groups; dead-letter; no retention requirement (outbox retains). NATS JetStream is the lighter alternative; Kafka only if broker-level replay is later required.
DR-3	Columnar analytics store timing	Not in phase 1; trigger when end-of-day aggregation exceeds a 2-hour window on the read replica	Provider-neutral candidates: ClickHouse (self-hosted) or DuckDB-on-object-storage for batch. The outbox makes this a rebuild.

#	Point	Default proposed	Criteria / consequence
DR-4	Role template drift policy	Notify and offer diff; no forced propagation	Alternative: forced propagation for a marked security-critical subset. Affects entity autonomy expectations (M-08).
DR-5	PIN policy	4–6 digits, entity-configurable; 5 attempts then 15-minute lockout; no reuse of last 3	Trade-off between counter speed and shared-login risk (N-03).
DR-6	Numerals in Sinhala and Tamil interfaces	Western Arabic digits everywhere	Native numerals would differ from receipts, tax documents and calculators used at counters; confirm with the usability prototype and real cashiers.
DR-7	MFA freshness window	12 hours; per-action for credit-limit change, control-price entry, external grants	Shorter windows slow finance work; longer windows weaken the control.
DR-8	Hash-chain anchor storage (phase 2)	Write-once object storage under a credential the application does not hold	Provider must offer object lock or equivalent; local hosting needs an equivalent (e.g. append-only export to a separate host).

12. Traceability

Service	Requirements and decisions
Identity	N-03, N-04, N-08, N-09; M-08 and ADR-18 (no downward admin); H-02 (cached permissions offline); Q-09 (helpdesk boundary); R-07 (OAuth2/OIDC API); FR-SEC-020 to 040; NFR security (OWASP ASVS L2, NIST credential guidance)
Permissions and roles	FR-SEC-010 (dynamic RBAC, templates); N-01, N-02, N-07 (segregation of duties); E-07, F-04 (value thresholds); A-11; M-05 (shop isolation); September 2026 single-staff shop decisions
Audit	FR-SEC-050; N-05 (immutable, event catalogue of doc 07 §3), N-06 (named reviewers), N-09 (drawer-open event); J-06, J-09; M-10; ADR-27; P2 hash chain and SIEM (doc 14 roadmap)
Internationalisation	FR-LOC-010 to 030; P-01 to P-09; H-09; B-14 (trilingual master data); usability directive (S-02, P-08)
Event model	ADR-10 (one writer); H-01, H-04; O-03 (near-real-time with freshness stamps); FR-REP-010; NFR availability and capacity (ADR-21, ADR-22); doc 18 P4, P6, P9, part E
Notifications delivery	R-08; FR-BIL statements; ADR-27; doc 20 M9
Configuration register	Doc 07 §4; P10 of doc 17; C-04, E-07, F-10, N-09; doc 32 §5.1
Scheduling	Doc 14 §1.3; ADR-39; B-I4, C-I2, E-I2; doc 31 ring promotion

End of document. Draft for design review; subject to revision after the decision requests in Section 8 are answered.